Flatness of representation counts for prime sequences, I: the sub-peak spectrum and the second-order ripple

Kentaro Amauchi*

August 8, 2026

Abstract

Let B be a finite set of odd primes and $r_B(m)$ the number of subsets of B summing to m. The companion paper reduces the asymptotic behaviour of the subset-sum landscape of B to a single quantity, the *flatness* ε_d, which measures the pointwise oscillation of r_B on a bounded window. This paper develops the Fourier-side input needed to bound ε_d: we determine the complete spectrum of sub-peaks of the characteristic product F_B, we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with m odd is exactly 0 whenever $m \in B$; for B a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \ldots, 40$. Key steps are formally verified in Lean 4.

1 Introduction

1.1 The quantity to be bounded

For a finite set B of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \qquad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta} \, d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant provided the *flatness*

$$\varepsilon_d = \max_{m, m' \in \mathrm{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where B_d is the subsequence of elements exceeding $2d$ and Win_d is a bounded window about $T/2$. Bounding ε_d is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

*Independent researcher. amkn.sub03@gmail.com, https://mathlab0911.github.io/

1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a\in B}\cos(a\theta/2)$, so that $|F_B| = 2^{|B|}|G_B|$, the natural per-element height at modulus q is

$$M(q) = \Big(\prod_{\substack{1\le r\le q\\ \gcd(r,q)=1}} |\cos(\pi r/q)|\Big)^{1/\varphi(q)}.$$

Theorem 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ (m odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction K_2 and phase drift ψ; Section 5 treats the main arc and identifies the first correction to the central density, resolving an unexplained constant of the companion paper. Section 6 reports the numerical verification.

1.3 Formal verification

Statements marked `name` are machine-checked in Lean 4 against Mathlib; the repository builds with no `sorry` and no additional axioms. The canon currently consists of 95 theorems.

2 The sub-peak spectrum

Theorem 2.1 (Cyclotomic evaluation)**.** *For every integer $q \ge 3$,*

$$M(q) = \tfrac{1}{2}\,|\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q-th cyclotomic polynomial.

Proof. Put $\zeta = e^{2\pi i/q}$. For $1 \le r \le q-1$ we have $|1+\zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r,q)=1}(x-\zeta^r)$, evaluating at $x = -1$ gives $\prod_{\gcd(r,q)=1}|1+\zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r,q)=1}|\cos(\pi r/q)| = 2^{-\varphi(q)}|\Phi_q(-1)|$. Take $\varphi(q)$-th roots. $\square$

Lemma 2.2 (Classical values)**.** *For $q \ge 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with p an odd prime and $j \ge 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \ge 2$; and $\Phi_q(-1) = 1$ otherwise.*

Proof. We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if n has at least two distinct prime factors; for odd $n \ge 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even m, $\Phi_{2m}(x) = \Phi_m(x^2)$. If q is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with m odd then $\Phi_q(-1) = \Phi_m(1)$, which is p when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with m even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when m is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. $\square$

Theorem 2.3 (Extremality of modulus six)**.** $\sup_{q\ge3} M(q) = \sqrt{3}/2$*, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \le 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.*

Proof. By Theorem 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in j with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2}\, p^{1/(p^{j-1}(p-1))}$; the exponent decreases in j, so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2}p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \cdots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2}3^{1/6} \approx 0.6005$. Combining the three families gives the statement. □

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

3 The vanishing family and the surviving spectrum

Lemma 3.1 (Exact vanishing)**.** *Let $q = 2m$ with m odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

Proof. The factor of F_B indexed by m is $1 + e^{i\pi} = 0$. □

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of m, so if B is a set of primes and $m > 1$ then B meets that class if and only if m itself lies in B. This gives a complete description.

Corollary 3.2 (Surviving spectrum)**.** *Let B_d be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly*, provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod 6$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.*
2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly*, for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd a. This is the Fourier form of the modulus-4 obstruction of the companion paper.*
3. $|G_{B_d}(2\pi/(2p))| = 0$ *exactly, for every prime p with $2d < p \leq x$.*
4. *For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.*
5. *For $q = 2m$ with m odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2}3^{1/6} + o(1))^b$.*
6. *For q odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is 0 by parity.*

Remark 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed d, only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes ≥ 5; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

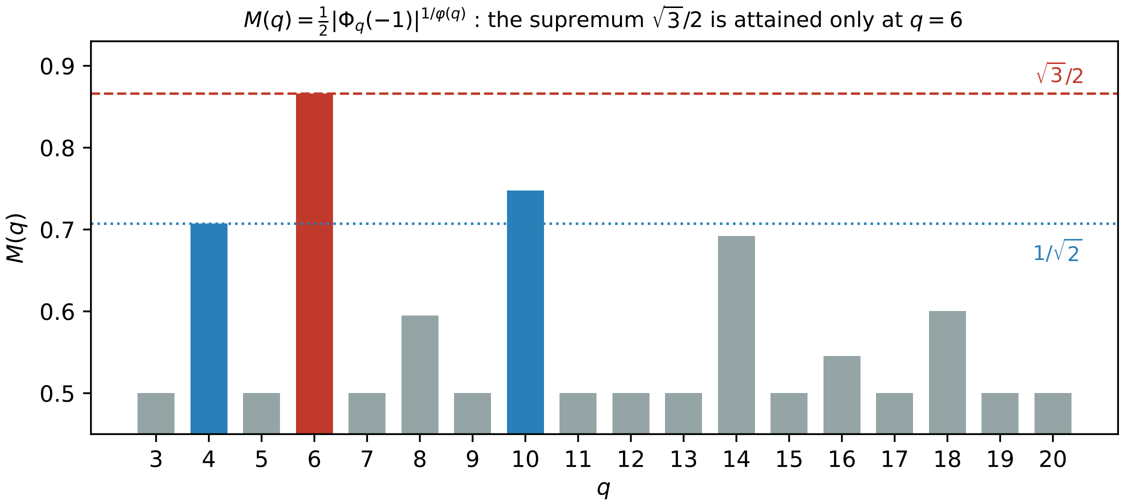

Figure 1: The per-element sub-peak height $M(q)$. The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

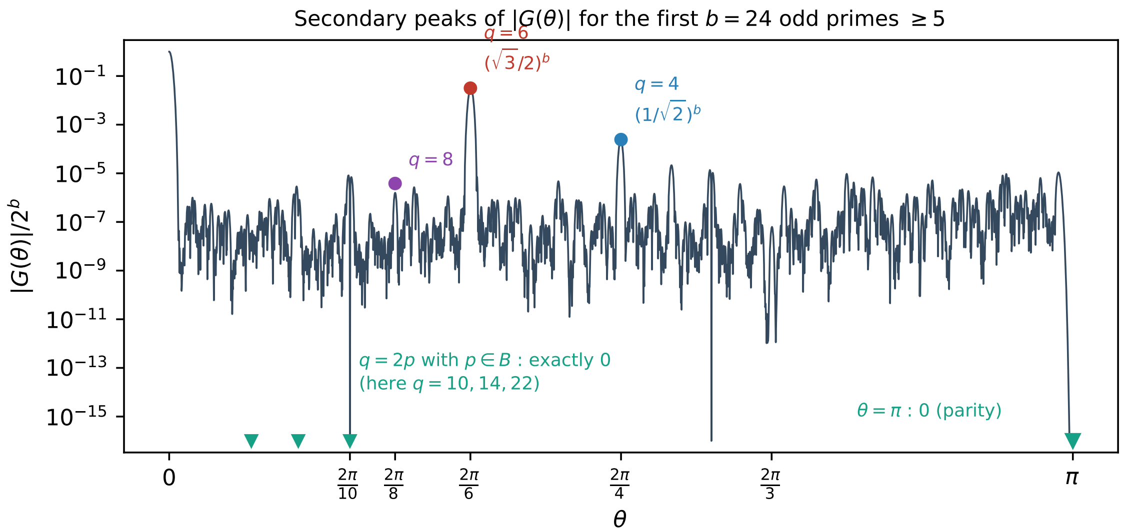

Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes ≥ 5. The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1); $\theta = \pi$ vanishes by parity.

4 The modulus-six ripple to second order

Throughout this section B is a set of b odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a\in B} a^2, \quad S_4 = \sum_{a\in B} a^4, \quad \Delta = \sum_{a\in B} \tau_a a, \quad \Delta_3 = \sum_{a\in B} \tau_a a^3,$$

where $\tau_a = +1$ if $a \equiv 1 \pmod 6$ and $\tau_a = -1$ if $a \equiv 5 \pmod 6$, and let c_1, c_5 be the corresponding class counts. Write $\mathrm{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

Lemma 4.1 (Local data at $\theta = \pi/3$)**.** *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log|G_B(\pi/3 + t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

Moreover $\frac{d}{d\theta}\arg(1 + e^{ia\theta}) = a/2$ exactly, so the phase of F_B is linear along the arc and all curvature lives in $|G_B|$.

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_mul_div_six`.)

Proposition 4.2 (Second-order ripple formula)**.** *For m in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\mathrm{Main}(m)} - 1 = 2\Big(\frac{\sqrt{3}}{2}\Big)^{b+1} K_2 \cos\Big(\frac{\pi}{6}(2m - (c_1 - c_5)) + \psi(\nu)\Big) + E,$$

where

$$K_2 = \exp\Big[\frac{\Delta^2}{8S_2} - \frac{3}{8}\frac{S_4}{S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^3} - \frac{3S_4\Delta^2}{16S_2^3}\Big], \qquad \psi(\nu) = -\frac{\sqrt{3}}{2}\,\nu\,\frac{\Delta + \Delta_3/S_2}{S_2}.$$

Derivation. Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L + i\nu)^2/(2V_6) + C_3(L + i\nu)^3/(6V_6^3) + \cdots$, the Gaussian-fluctuation factor $-\frac{1}{2}\log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \cdots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives K_2 and ψ. The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b + 1$. □

Remark 4.3 (The two branches). The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \bmod 6$; the term $-\frac{3}{8}S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \geq 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \leq 47$ and $K_2 < 1$.

Remark 4.4 (Error policy). The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in Δ, of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

5 The main arc and the central density

The main arc contributes the smooth part of r_B. The leading term is the local central limit theorem; what matters for the companion paper's absolute formula is the *first correction* to it.

Proposition 5.1 (Central density with first correction). *Let B be a set of b odd integers, $V_0 = S_2/4$, and let m be at the centre $T/2$. Then*

$$r_B(m) = \frac{2^b}{\sqrt{2\pi V_0}}\Big(1 - \frac{S_4}{4S_2^2} + O((S_4/S_2^2)^2)\Big).$$

Derivation. Write $\log G_B(\theta) = \sum_{a\in B} \log\cos(a\theta/2) = -\frac{V_0}{2}\theta^2 - \frac{S_4}{192}\theta^4 + O(S_6\theta^6)$, using $\log\cos u = -u^2/2 - u^4/12 + O(u^6)$ and $u = a\theta/2$. Since G_B is even (Lean: `G_even`), all odd-order terms vanish identically, so the first correction is the quartic one. Integrating $e^{-V_0\theta^2/2}(1 - \frac{S_4}{192}\theta^4)$ over the main arc and using $\int \theta^4 e^{-V_0\theta^2/2} = 3\sqrt{2\pi}V_0^{-5/2}$ gives the stated factor $1 - \frac{S_4}{192}\cdot 3V_0^{-2} = 1 - \frac{S_4}{4S_2^2}$. □

Remark 5.2 (The absolute formula of the companion paper). The companion paper predicts the number of strict local minima by $\mathrm{lm} \approx \Gamma \cdot 2^b/\sqrt{2\pi V_0}$, i.e. by replacing the ground-state count by its Gaussian estimate. The measured ratio of the two sides sits systematically below 1 — about 0.97 — with no explanation offered there. Proposition 5.1 identifies the deficit as $S_4/(4S_2^2)$, which at $k = 20$ equals 0.0246. Section 6.4 verifies this.

6 Numerical verification

All computations below are exact integer dynamic programming for r_B, followed by a discrete Fourier extraction; logs are included in the repository.

6.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

Δ_3^2 coefficient	$S_4\Delta^2$ term	branch A mean	branch B mean
45/24	no	0.00061	0.00736
45/24	yes	0.00060	0.00400
5/24	no	0.00073	0.00411
5/24	yes	0.00072	**0.00073**

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \ldots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

On branch B the individual values of measured$/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$. The omitted contributions are dominated by the class-independent tail $O((S_4/S_2^2)^2)$, and S_4/S_2^2 decreases monotonically over the range, from 0.1111 at $k = 18$ to 0.0507 at $k = 40$; hence the residual is largest at the smallest k. That the Δ-dependent quantity Δ_3^2/S_2^3 is in fact largest at $k = 32$ ($2.70 \cdot 10^{-3}$, against $1.04 \cdot 10^{-3}$ at $k = 18$) without producing an excess residual there is corroborating evidence that the Δ-dependence is already absorbed into K_2.

6.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \ldots, 40$, only α is significant, at every window width tested: $t(\alpha)$ ranges from $+4.26$ to $+5.16$, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model measured $= c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths $24, 48, 96$ and 192, with R^2 from 0.978 to 0.986; restricting to $|\Delta| \geq 17$ raises R^2 to 0.993. We conclude that ψ as stated is correct, with the deficit in c a finite-window effect.

Remark 6.1 (Extraction). The amplitude is extracted from a narrow window of fixed half-width 24, i.e. eight full periods of the modulus-6 ripple, which over $k = 18, \ldots, 40$ is between 0.08 and 0.30 standard deviations; both requirements — enough periods, and narrower than σ — hold throughout. Wide windows corrupt the Gaussian reference. As a check, at $k = 18$ the ratio measured$/K_2$ is 0.9988, 0.9974, 1.0034 and 1.0065 at half-widths $12, 16, 24$ and 32, so the window choice moves the residual by less than 1%. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

6.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level d: a peak at $q = 2p$ is dark exactly when p is still present in B_d, and lights up as soon as the truncation removes p. All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

d	removed primes	$q = 6$	$q = 10$	$q = 14$	tallest
1	—	0	0	0	$q = 4$
2	3	$3.66 \cdot 10^{-2}$	0	0	$q = 6$
3	3, 5	$4.22 \cdot 10^{-2}$	$1.03 \cdot 10^{-3}$	0	$q = 6$
4	3, 5, 7	$4.88 \cdot 10^{-2}$	$1.75 \cdot 10^{-3}$	$4.33 \cdot 10^{-4}$	$q = 6$

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for A the first 24 odd primes. Entries equal to 0 are exact zeros.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every d, confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and 2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter. This is to be expected, since the error in the prime number theorem in arithmetic progressions decreases only logarithmically, so at $b = 21, \ldots, 31$ the $o(1)$ in items (iv)–(v) of Corollary 3.2 has not yet become small.

6.4 The central density

We test Proposition 5.1 using only ground-state counts. Write $\lambda(B) = \deg(B)\sqrt{2\pi V_0}/2^b - 1$ for the measured relative deficit, where $\deg(B)$ is the number of subsets attaining the minimum of $|\sum S - \lfloor T/2 \rfloor|$. Proposition 5.1 predicts $\lambda \approx -S_4/(4S_2^2)$.

Over an ensemble of 100 random odd sequences per size (drawn from the same range as the primes), the mean absolute difference is 0.0171 at $k = 16$, 0.0079 at $k = 18$ and 0.0038 at $k = 20$; the ensemble is noisier than the primes because deg is a small integer subject to arithmetic fluctuation.

The consequence for Remark 5.2 is direct. Writing $\rho = \mathrm{lm}/(\Gamma \cdot 2^b/\sqrt{2\pi V_0})$ for the ratio in the absolute formula, the ensemble mean of ρ is 0.9462, 0.9621, 0.9672 and 0.9716 at $k = 12, 16, 18, 20$, while the mean of $\rho/(1 - S_4/(4S_2^2))$ is 0.9812, 0.9888, 0.9912 and 0.9933.

k	deg	measured λ	predicted	difference
16	394	-0.024764	-0.031338	$+0.006573$
18	1290	-0.025573	-0.027754	$+0.002181$
20	4344	-0.024036	-0.024629	$+0.000594$
24	51068	-0.021467	-0.020830	-0.000637
28	634568	-0.017663	-0.017574	-0.000090

Table 3: Central-density deficit for the first k odd primes. The agreement is within 0.5% for $k \geq 18$ and improves to 0.01% at $k = 28$.

The correction removes about three quarters of the deficit, and the share it removes grows with k. We therefore record the systematic part of the 0.97 as explained; the residual 0.7% at $k = 20$ is a separate and smaller effect, still decreasing.

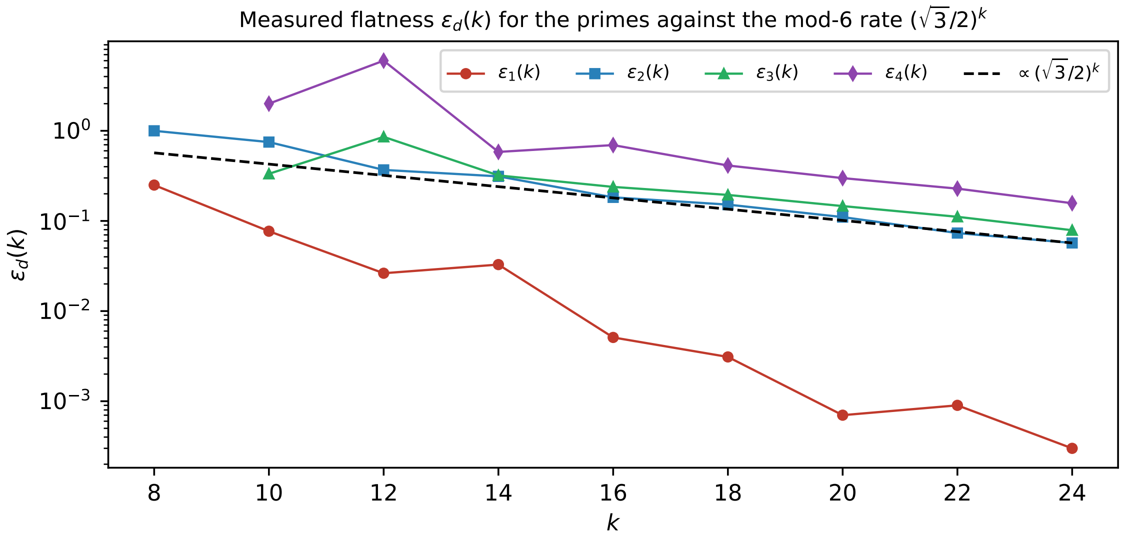

Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1).

7 A free Gaussian envelope at every peak

The following is elementary but useful: every sub-peak carries an envelope as sharp as the main arc, with no arithmetic input at all.

Proposition 7.1 (Curvature bound). *Let B be any finite set of positive reals and $V_0 = \frac{1}{4}\sum_{a\in B} a^2$. On any interval where G_B has no zero,*

$$\frac{d^2}{d\theta^2} \log |G_B(\theta)| = -\sum_{a\in B} \frac{a^2}{4} \sec^2\!\Big(\frac{a\theta}{2}\Big) \;\leq\; -V_0,$$

because $\sec^2 \geq 1$. Consequently $\log |G_B|$ is concave there with curvature at least V_0, and if θ^* is an interior local maximum then

$$|G_B(\theta)| \;\leq\; |G_B(\theta^*)|\, e^{-V_0(\theta-\theta^*)^2/2}$$

throughout that interval.

(Lean: `one_le_one_div_cos_sq`, `term_curvature_ge` and `curvature_ge_V0` formalise the inequality $V_0 \le \sum(a^2/4)\sec^2(a\theta/2)$; the passage from concavity to the quadratic bound is standard and not yet formalised.)

Two caveats matter in practice, and both are visible numerically.

Remark 7.2 (The centre is not $2\pi/q$). The envelope must be anchored at the true local maximum, which is displaced from $2\pi/q$ by $-h'(0)/h''(0)$ with $h = \log|G_B|$; by Lemma 4.1 this is $\asymp \Delta/S_2$ at $q = 6$. For the first 24 odd primes truncated at $d = 2$ the measured displacements are $+1.316 \cdot 10^{-5}$ ($q = 6$), $+3.795 \cdot 10^{-4}$ ($q = 4$) and $-2.097 \cdot 10^{-3}$ ($q = 8$), against predicted $+1.316 \cdot 10^{-5}$, $+3.800 \cdot 10^{-4}$ and $-2.021 \cdot 10^{-3}$. Anchoring at $2\pi/q$ instead makes the envelope fail, by $7.6 \cdot 10^{-6}$ at $q = 6$ and $4.7 \cdot 10^{-3}$ at $q = 4$ in the same data.

Remark 7.3 (Validity radius). The bound holds only inside the zero-free interval, whose half-width is $\approx \pi/(2\max B)$. Measured in units of the envelope scale $V_0^{-1/2} = 2/\sqrt{S_2}$, this is

$$\frac{\pi}{2\max B} \cdot \frac{\sqrt{S_2}}{2} \;\approx\; \frac{\pi}{4}\sqrt{k/3} \;\approx\; 0.45\sqrt{k},$$

so the envelope covers a growing number of standard deviations: 1.72 at $k = 16$, 2.08 at $k = 24$, 2.71 at $k = 40$ and 2.92 at $k = 48$, against the prediction $1.80, 2.21, 2.85, 3.12$.

8 What remains

The results above supply the arithmetic input for a circle-method treatment of ε_d. Three steps remain, and we state them explicitly.

Problem 8.1 (Main arc)**.** Make the local central limit theorem for r_B on the main arc effective, with an error term of the shape C/S_2 arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in b.

Problem 8.2 (Minor arcs)**.** Bound the contribution of θ away from the surviving peaks. By Corollary 3.2 it suffices to treat $q = 4$, $q = 6$, and a finite list depending on d.

Problem 8.3 (Assembly)**.** Combine the two to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \to 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for ε_d can be extracted from the aggregate residue-class spread. The route to Problem 8.3 must go through the upper bounds above.